

DHCP Snooping

- Topics covered:
 - What Is DHCP Snooping?
 - How Does It Work?
 - What Attacks Does It Prevents?
 - DHCP Snooping Configuration

What Is DHCP Snooping?

DHCP Snooping

This section introduces DHCP Snooping, how it works, the attacks it prevents, and its basic operation on Cisco switches.

What is DHCP Snooping?

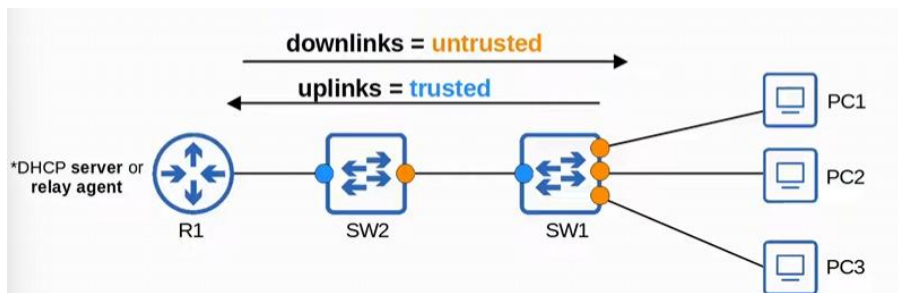
DHCP Snooping is a switch security feature used to **filter DHCP messages received** on untrusted ports.

1. It only affects **DHCP traffic**.
2. Non-DHCP traffic is forwarded normally.
3. All switch ports are **untrusted by default**.
4. Administrators must manually configure which ports are **trusted**.

Typically:

- **Uplink ports** → Trusted
- **Downlink ports** → Untrusted

Network Layout



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1. R1 acts as either a DHCP server or DHCP relay agent.
2. End hosts obtain IP addresses from R1 using DHCP.

Downlink Ports (Untrusted)

These interfaces point toward end-user devices.

Reasons they remain untrusted:

- Administrators do not have direct control over user devices.
- A malicious user could launch DHCP-based attacks.
- Therefore, these ports should remain in their default untrusted state.

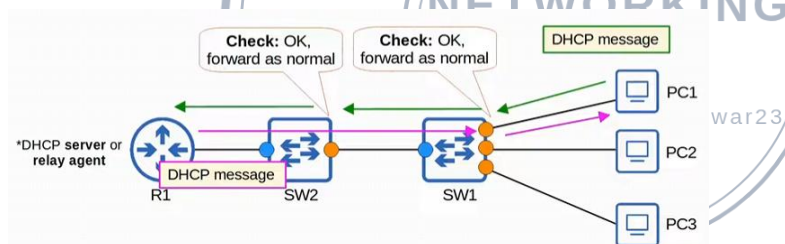
Uplink Ports (Trusted)

These interfaces point toward network infrastructure devices controlled by the administrator.

Because infrastructure devices are trusted, DHCP Snooping does not inspect DHCP traffic received on these interfaces.

DHCP Snooping Operation

Example 1: Valid DHCP Message



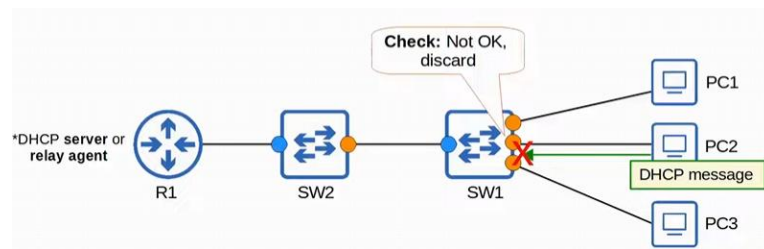
Process:

1. PC1 sends a DHCP message (for example, a Discover message).
2. SW1 receives it on an untrusted port.
3. SW1 inspects the message.
4. The message is determined to be valid and is forwarded.
5. SW2 receives the message on an untrusted interface.
6. SW2 inspects it and determines it is valid.
7. SW2 forwards it to R1.
8. R1 sends a reply.
9. The reply passes through SW2 and SW1 without inspection.

The reply is not inspected because it arrives on **trusted ports**, and the switches assume the message is valid.

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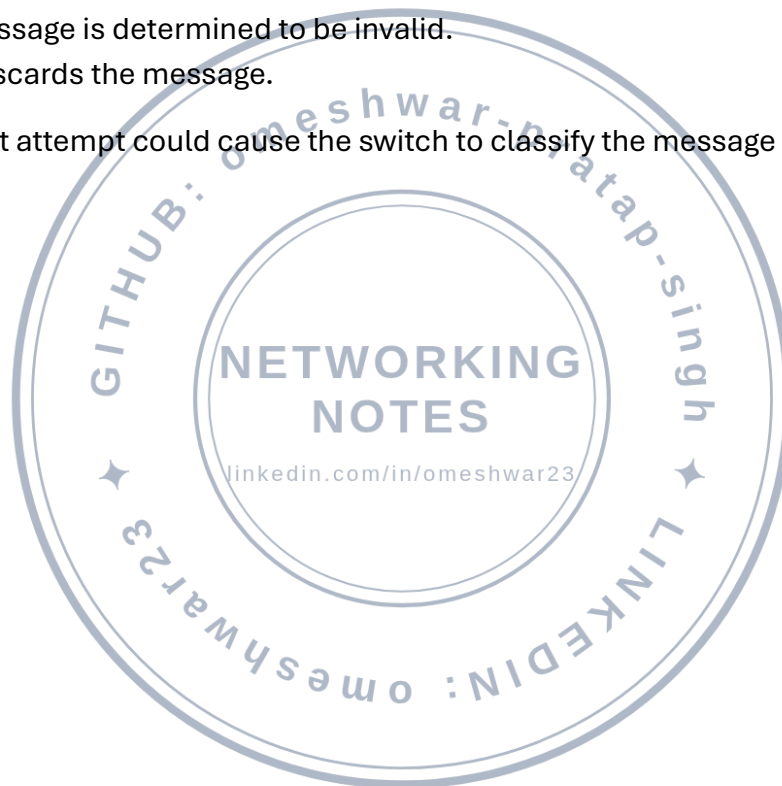
Example 2: Invalid DHCP Message



Process:

1. PC2 sends a DHCP message.
2. SW1 receives it on an untrusted port.
3. SW1 inspects the message.
4. The message is determined to be invalid.
5. SW1 discards the message.

A DHCP exploit attempt could cause the switch to classify the message as invalid and drop it.



What Attacks Does It Prevents?

1. DHCP Starvation (DHCP Exhaustion)

DHCP Starvation is a denial-of-service attack.

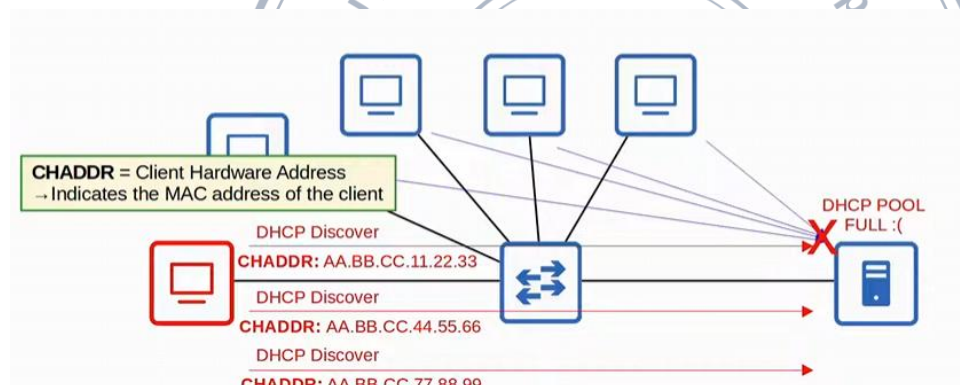
Purpose

The attacker floods the DHCP server with requests using spoofed MAC addresses until the DHCP address pool becomes exhausted.

As a result:

- The DHCP pool becomes full.
- Legitimate devices cannot obtain IP addresses.
- A denial of service occurs.

Attack Process



DHCP Pool Full, the attacker repeatedly sends DHCP Discover messages using fake MAC addresses.

CHADDR Field

CHADDR = Client Hardware Address = Client MAC Address

DHCP includes a field called **CHADDR (Client Hardware Address)** that identifies the client's MAC address.

Why is CHADDR needed?

1. If the DHCP server is on the local network, the source MAC address of the frame could identify the client.
2. If a DHCP relay agent forwards the message to a remote DHCP server, the frame's source MAC address will no longer be the client's MAC address.
3. Therefore, the CHADDR field is used to identify the client.

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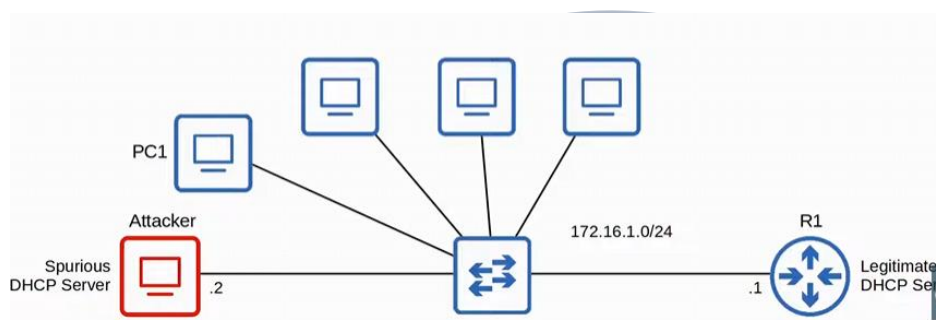
The attacker abuses this by sending many Discover messages containing different CHADDR values.

Eventually, the DHCP server's pool is exhausted, preventing legitimate clients from obtaining addresses.

2. DHCP Poisoning

DHCP Poisoning is similar in purpose to ARP Poisoning and can be used for a **man-in-the-middle attack**.

Network Scenario



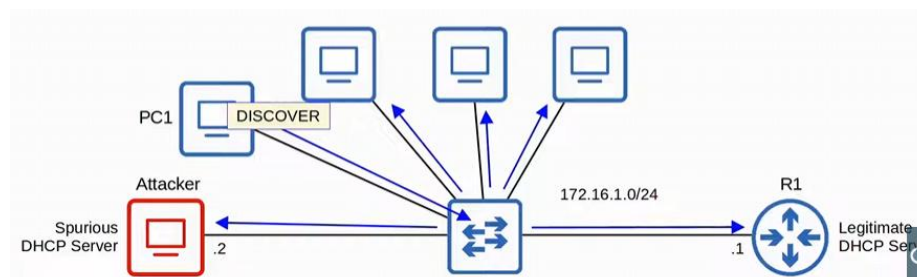
- The attacker runs DHCP server software.
- This creates a **spurious (illegitimate) DHCP server**.
- R1 is the legitimate DHCP server and default gateway.

Both servers respond to client DHCP Discover messages.

Clients typically accept the **first DHCP Offer** they receive.

If the legitimate DHCP server is in a remote network and traffic must pass through a DHCP relay agent, the attacker's Offer is very likely to arrive first because it is local.

Step 1: Client Sends Discover

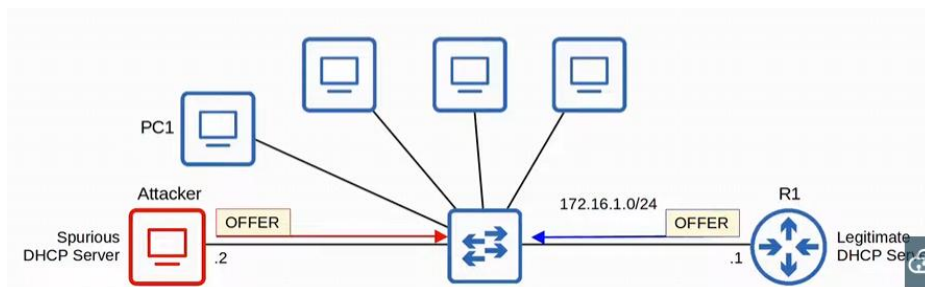


PC1 broadcasts a DHCP Discover message.

Both the attacker and R1 receive it.

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Step 2: Both Servers Send Offers

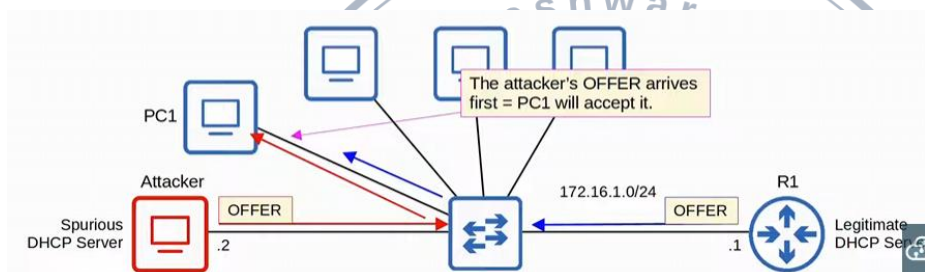


Both servers send DHCP Offer messages.

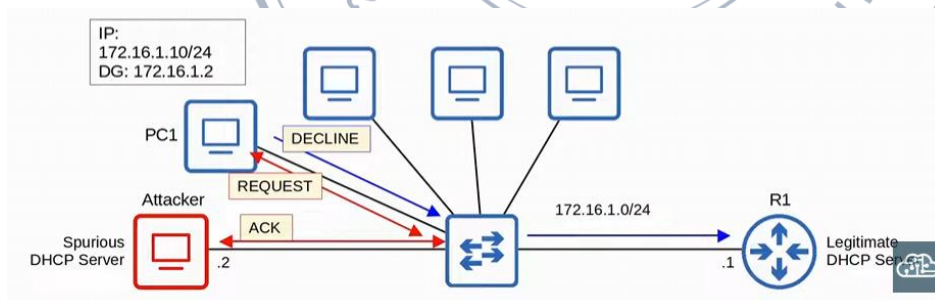
The attacker's Offer arrives first.

Therefore, PC1 accepts the attacker's Offer.

Step 3: DHCP Completion



- PC1 accepts the attacker's Offer.
- PC1 sends a DECLINE message to R1.
- DHCP completes with the attacker.

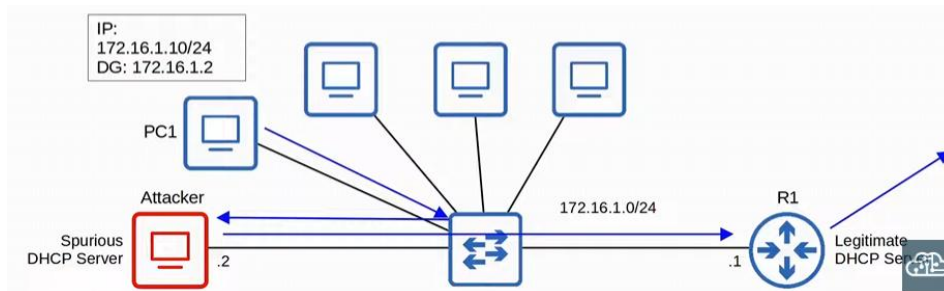


PC1 receives:

- IP Address: **172.16.1.10/24**
- Default Gateway: **172.16.1.2** (the attacker)

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Step 4: Man-in-the-Middle Traffic Flow



Because the attacker is configured as the default gateway:

1. PC1 sends external traffic to the attacker.
2. The attacker can inspect the traffic.
3. The attacker can modify the traffic.
4. The attacker forwards it to the legitimate default gateway (R1).
5. R1 sends it toward the destination.

Since the traffic still reaches its destination, PC1 may not realize that its communications are being intercepted.

DHCP Server Messages vs DHCP Client Messages

When DHCP Snooping filters traffic, it treats **DHCP server messages** and **DHCP client messages** differently.

DHCP Server Messages

These messages are sent by DHCP servers:

1. **OFFER** – Used in the DHCP D-O-R-A process.
2. **ACK** – Used in the DHCP D-O-R-A process to acknowledge a request.
3. **NAK** – Opposite of ACK; used to reject a client's REQUEST message.

DHCP Client Messages

These messages are sent by DHCP clients:

1. **DISCOVER** – Used in the D-O-R-A process.
2. **REQUEST** – Used in the D-O-R-A process.
3. **RELEASE** – Indicates the client no longer requires its IP address.
4. **DECLINE** – Rejects an IP address offered by a DHCP server.

Understanding which messages originate from DHCP servers and which originate from DHCP clients is essential for understanding DHCP Snooping.

DHCP Snooping Configuration

DHCP Snooping Filtering Process

Step 1: Message Received on a Trusted Port

DHCP Message → Trusted Port → Forward Normally (No Inspection)

If a DHCP message arrives on a trusted port:

1. The switch forwards it normally.
2. No inspection occurs.
3. Messages received on trusted ports are not dropped.

Step 2: Message Received on an Untrusted Port

DHCP Message → Untrusted Port → Inspect Message

If a DHCP message arrives on an untrusted port, the switch inspects it and determines how to handle it.

DHCP Server Messages on Untrusted Ports

OFFER / ACK / NAK → Untrusted Port → X Discard

If an OFFER, ACK, or NAK message is received on an untrusted port:

- The switch immediately discards it.
- No additional checks are performed.

DHCP servers should be connected through trusted ports. Therefore, DHCP server messages arriving on untrusted ports are considered invalid.

DHCP Client Message Checks (DISCOVER and REQUEST Validation)

For DISCOVER and REQUEST messages:

Frame Source MAC

| Compare

DHCP CHADDR Field

The switch compares:

1. Frame source MAC address
2. DHCP Client Hardware Address (CHADDR)

Results:

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- Match → Forward the frame
- No Match → Discard the frame

This helps protect against attacks that spoof the DHCP client hardware address field.

However, it is not perfect because an attacker could also spoof the frame's source MAC address.

RELEASE and DECLINE Validation

For RELEASE and DECLINE messages:

Source IP Address+Receiving Interface

| Compare

DHCP Snooping Binding Table

The switch verifies whether:

- Source IP address
- Receiving interface

match an entry in the DHCP Snooping Binding Table.

Results:

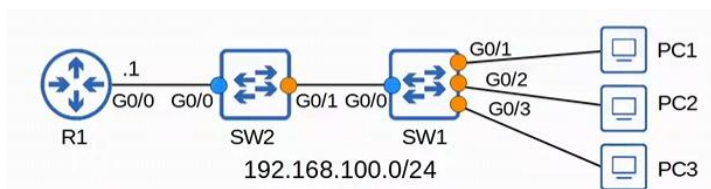
- Match → Forward
- No Match → Discard

DHCP Snooping Binding Table

When DHCP Snooping is enabled and a client successfully receives an IP lease, the switch creates an entry in the DHCP Snooping Binding Table.

Basic DHCP Snooping Configuration

Network Topology



SW2 Configuration

```
SW2(config)#ip dhcp snooping
SW2(config)#ip dhcp snooping vlan 1
SW2(config)#no ip dhcp snooping information option
SW2(config)#interface g0/0
SW2(config-if)#ip dhcp snooping trust
```

I will explain this later!

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Explanation

1. ip dhcp snooping: Enables DHCP Snooping globally on the switch.
2. ip dhcp snooping vlan 1: Enables DHCP Snooping for VLAN 1.

DHCP Snooping must be:

- Enabled globally.
- Enabled on each required VLAN.

In this network, VLAN 1 is the only VLAN.

3. no ip dhcp snooping information option: Mentioned here and explained later.
4. ip dhcp snooping trust: Configures G0/0 as a trusted port.

G0/0 connects toward R1, the DHCP server.

SW1 Configuration

```
SW1(config)#ip dhcp snooping
SW1(config)#ip dhcp snooping vlan 1
SW1(config)#no ip dhcp snooping information option
SW1(config)#interface g0/0
SW1(config-if)#ip dhcp snooping trust
```

Explanation

The same configuration is applied on SW1.

1. G0/0 is trusted.
2. It points toward R1, the DHCP server.

Viewing the DHCP Snooping Binding Table

```
SW1#show ip dhcp snooping binding
MacAddress      IpAddress      Lease(sec)  Type           VLAN  Interface
-----
0C:29:2F:18:79:00  192.168.100.10  86294      dhcp-snooping  1     GigabitEthernet0/3
0C:29:2F:90:91:00  192.168.100.11  86302      dhcp-snooping  1     GigabitEthernet0/1
0C:29:2F:67:E9:00  192.168.100.12  86314      dhcp-snooping  1     GigabitEthernet0/2
Total number of bindings: 3
```

Information Stored

The table records: MAC address, IP address, Lease time, VLAN number, Interface ID

The RELEASE and DECLINE checks use this table.

RELEASE/DECLINE messages will be checked to make sure their IP address/interface ID match the entry in the DHCP snooping table.

The switch verifies that the IP address and interface match the binding-table entry before forwarding those messages.

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This prevents attackers from sending RELEASE or DECLINE messages on behalf of other devices and causing the DHCP server to believe those devices no longer require their IP addresses.

DHCP Snooping Rate Limiting

Another DHCP Snooping feature is rate limiting.

The switch can limit how many DHCP messages are allowed to enter an interface.

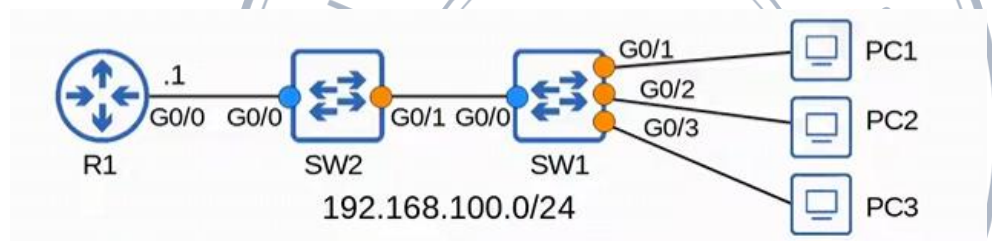
If the configured rate is exceeded: The interface becomes **err-disabled**.

The interface can later be restored by:

1. shutdown
2. no shutdown

or through err-disable recovery.

Network Topology for Rate Limiting



Rate-Limit Configuration

```
SW1(config)#interface range g0/1 - 3
SW1(config-if-range)#ip dhcp snooping limit rate 1
```

Explanation

The command: `ip dhcp snooping limit rate 1`

limits DHCP traffic to:

1 DHCP message per second

If more than one DHCP message per second arrives, the interface is disabled.

The value of 1 message per second is intentionally very low. In a normal network, such a setting would cause interfaces to become disabled even during legitimate DHCP activity.

Interface Becomes Err-Disabled

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```
*Jun  5 13:15:14.180: %DHCP_SNOOPING-4-DHCP_SNOOPING_ERRDISABLE_WARNING: DHCP Snooping received 1 DHCP packets on interface Gi0/1
*Jun  5 13:15:14.181: %DHCP_SNOOPING-4-DHCP_SNOOPING_RATE_LIMIT_EXCEEDED: The interface Gi0/1 is receiving more than the threshold set
*Jun  5 13:15:14.182: %PM-4-ERR_DISABLE: dhcp-rate-limit error detected on Gi0/1, putting Gi0/1 in err-disable state
*Jun  5 13:15:15.185: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
*Jun  5 13:15:16.190: %LINK-3-UPDOWN: Interface GigabitEthernet0/1, changed state to down
```

Result

These log messages indicate:

1. Gi0/1 received DHCP traffic.
2. The configured DHCP rate limit was exceeded.
3. A DHCP rate-limit violation was detected.
4. Gi0/1 was placed into an err-disabled state.
5. The interface transitioned to the down state.

DHCP Snooping Rate Limiting Recovery

Previously, an interface was placed into an **err-disabled** state because it exceeded the configured DHCP rate limit.

Instead of manually re-enabling the interface, **errdisable recovery** can be configured.

Configuration

```
SW1(config)#errdisable recovery cause dhcp-rate-limit
```

```
SW1#show errdisable recovery
```

ErrDisable Reason	Timer Status
arp-inspection	Disabled
bpduguard	Disabled
channel-misconfig (STP)	Disabled
dhcp-rate-limit	Enabled
dtp-flap	Disabled
gbic-invalid	Disabled
inline-power	Disabled

! [output omitted due to length]

Rate-limiting can be very useful to protect against DHCP exhaustion attacks.

```
Timer interval: 300 seconds
```

```
Interfaces that will be enabled at the next timeout:
```

Interface	Errdisable reason	Time left(sec)
Gi0/1	dhcp-rate-limit	293

Explanation

- errdisable recovery cause dhcp-rate-limit enables automatic recovery for DHCP rate limit violations.
- show errdisable recovery confirms that recovery is enabled for the **dhcp-rate-limit** cause.
- The output also shows that **Gi0/1** is waiting for the recovery timer to expire before being re-enabled.

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Purpose of DHCP Rate Limiting

DHCP rate limiting is useful for protecting against **DHCP exhaustion (starvation) attacks**.

Previously, it was shown that an attacker can:

- Spoof the frame's source MAC address.
- Spoof the DHCP client's hardware address (CHADDR).

This allows the attacker to bypass the DISCOVER and REQUEST checks performed by DHCP Snooping.

However, DHCP rate limiting provides additional protection.

If an attacker floods the network with large numbers of DHCP DISCOVER messages:

Attacker

|
| DHCP DISCOVER
| DHCP DISCOVER
| DHCP DISCOVER
| DHCP DISCOVER

v

[SW1 Interface]

|
X

Err-Disabled

Once the configured rate is exceeded:

1. The interface is disabled.
2. The attacker can no longer continue flooding the DHCP server.

This makes DHCP rate limiting a very useful DHCP Snooping feature.



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DHCP Snooping Information Option (Option 82)

Earlier, the following command was configured:

```
no ip dhcp snooping information option
```

To understand why, it is necessary to understand **Option 82**.

What is Option 82?

Option 82, also called the **DHCP Relay Agent Information Option**, is one of the DHCP options available within DHCP messages.

It provides additional information such as:

- Which DHCP relay agent received the client's message.
- Which interface received it.
- Which VLAN received it.

DHCP relay agents can add Option 82 when forwarding DHCP messages to a remote DHCP server.

Default Cisco Switch Behaviour

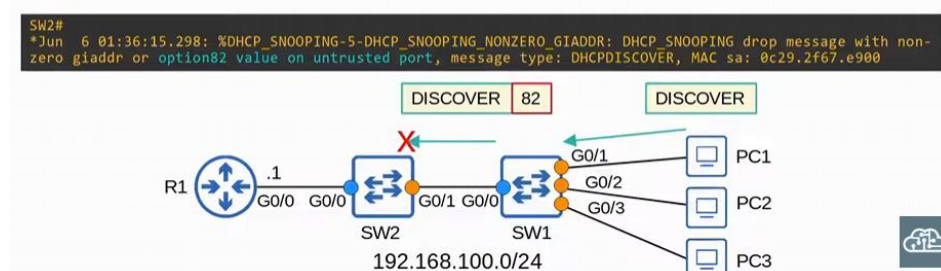
When DHCP Snooping is enabled:

1. Cisco switches add Option 82 to DHCP messages received from clients by default.
2. This occurs even when the switch is **not** acting as a DHCP relay agent.

This behavior can create problems.

Problem 1: Switch Drops a Message Containing Option 82

Network Topology



PC1 sends a DHCP DISCOVER message.

SW1:

1. Receives the message.
2. Adds Option 82.
3. Forwards it to SW2.

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SW2 receives the message on an untrusted port.

Because the message contains Option 82, SW2 drops it.

CLI Output

SW2#

The log indicates that the DHCP DISCOVER message was dropped because it contained Option 82.

The default behavior works well when a switch is functioning as a Layer 3 DHCP relay agent, but not in this network design.

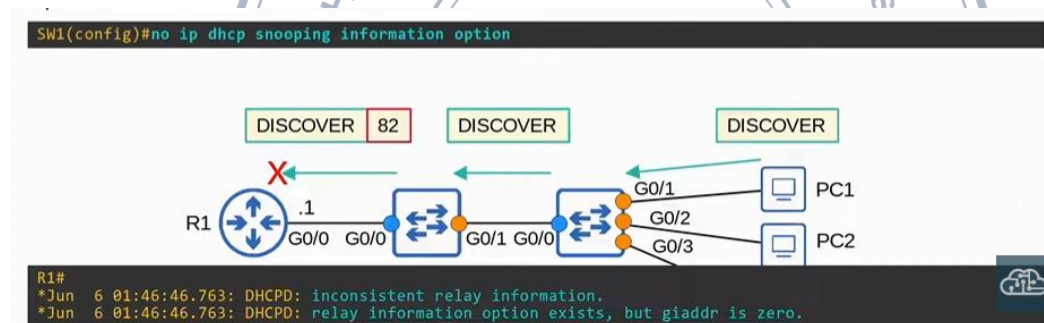
This is why the following command was configured:

no ip dhcp snooping information option

Problem 2: R1 Drops the Message

Assume the command was configured on SW1 but **not yet configured on SW2**.

Configuration



Process:

1. PC1 sends a DISCOVER message.
2. SW1 forwards it without adding Option 82.
3. SW2 adds Option 82.
4. SW2 forwards the message to R1.
5. R1 drops the message.

CLI Output Explanation

R1 considers the relay information inconsistent.

The message contains Option 82, but it was not sent by a DHCP relay agent.

Therefore, R1 drops the message.

The log message's **"inconsistent relay information"** indicates exactly this condition.

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Solution

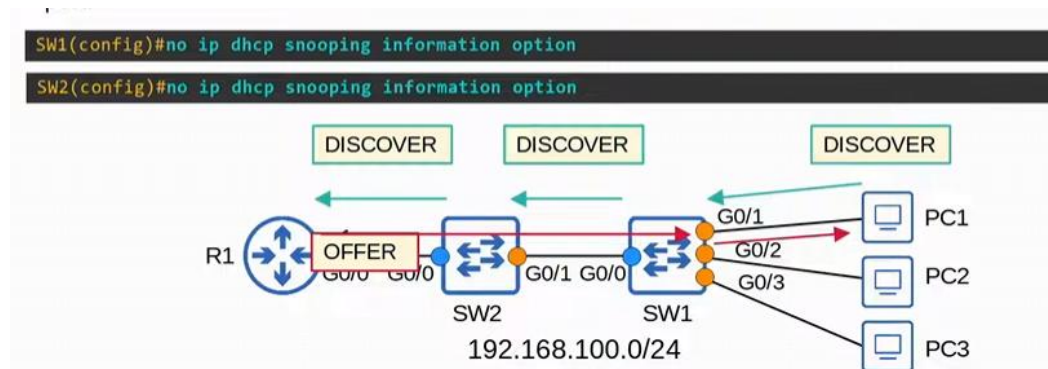
The command is configured on both switches.

Configuration

SW1(config)#no ip dhcp snooping information option

SW2(config)#no ip dhcp snooping information option

Result



As a result:

1. PC1 sends a DISCOVER message.
2. SW1 forwards it normally.
3. SW2 forwards it normally.
4. R1 accepts the message.
5. R1 responds with an OFFER.
6. DHCP communication proceeds normally.

Command Summary

```
SW1(config)# ip dhcp snooping

SW1(config)# ip dhcp snooping vlan vlan-number

SW1(config)# errdisable recovery cause dhcp-rate-limit

SW1(config)# no ip dhcp snooping information option

SW1(config-if)# ip dhcp snooping trust

SW1(config-if)# ip dhcp snooping limit rate packets-per-second

SW1# show ip dhcp snooping binding
```

